

Vivek Rallapally

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SUMMARY

Full-stack engineer with hands-on experience building end-to-end systems using Next.js, React.js, TypeScript, FastAPI, and Python. Proficient in real-time communication, async task processing, LLM integration, and production ML pipelines — comfortable owning both the frontend and the backend.

EDUCATION

MLR Institute of Technology	Telangana, India
Bachelors in Computer Science Engineering - AIML CGPA: 7.81	2021 - 2025
Excellencia Junior College	Telangana, India
Telangana State Board of Intermediate Education: 85.2%	2019 - 2021
Sherwood Public School	Telangana, India
Indian Certificate of Secondary Education: 73%	2007 - 2019

SKILLS SUMMARY

Languages: Java, Python, HTML, CSS, JavaScript, TypeScript
Frameworks and Libraries: React.js, TensorFlow, scikit-learn, PyTorch, Next.js, Ollama, Flask, FastAPI, Redis, Celery, WebSockets
Tools: Android Studio, Visual Studio, Figma, Jupyter, Google Colab, Git, Docker
Cloud: AWS

PROJECTS

[Smart Traffic Management System Using YOLOv11 and GBML DQN](#) | Reinforcement Learning

- Trained a GBML-DQN agent using YOLOv11 for vehicle detection in a SUMO simulation, reducing average queue length from ~55 to ~35 vehicles (~36% reduction) across 600 training episodes.
- Implemented real-time vehicle counting using YOLOv11 object detection on SUMO-simulated video streams, feeding live density estimates into the DQN state space.

SED (System Execution Dashboard) | AI Systems / Full-Stack

- Built a local AI-powered laptop automation system using Python, FastAPI, Next.js, and TypeScript with real-time WebSocket communication between frontend and backend.
- Integrated a local LLM (Qwen3 via Ollama) with tool calling and chaining, enabling natural language commands to execute OS-level automation including app control, system monitoring, and web search.
- Implemented async task processing with Redis and Celery, supporting concurrent automation jobs with persistent conversation context across sessions.

[NoPhoneDrive](#) | Android App

- Built an Android app using Java that detects driving state via accelerometer and GPS speed thresholds, automatically restricting phone usage above 20 km/h.

[XTree](#) | Web Technologies

- Developed a Chrome extension using JavaScript that scrapes and exports all anchor tags from any webpage to a CSV file, reducing manual data collection from ~10 minutes to under 5 seconds per page.

PXP-Fraudnet | Machine Learning

- Architected a 3-layer feedforward neural network for fraud detection in payment transactions.
- Designed a React.js dashboard displaying real-time fraud risk scores, transaction timelines, and model confidence metrics for analyst review.

[Translator](#) | Natural Language Processing

- Developed a text translation model to convert multilingual text into English.
- Applied tokenization, stopword removal, and sequence padding on multilingual corpora to preprocess input for a seq2seq translation model.

WORK EXPERIENCE

Intern at PXP February 2025 - May 2025

- React.js, PyTorch
- Constructed a PyTorch-based fraud detection model trained on 10,000+ transaction records, achieving 94% classification accuracy and a 25% increase in recall while maintaining under 1% FPR.
 - Exposed ML model predictions via a REST API integrated with a React.js dashboard, enabling real-time transaction risk visualization for end users.

- Delivered a production-ready ML pipeline covering data preprocessing, model training, evaluation, and frontend integration within a 3-month internship.

Virtual Intern for Google AIML | EduSkills AICTE COHORT-8

April 2024 - June 2024

TensorFlow, OpenCV, Google Colab Notebook

- Completed 3 end-to-end ML projects over 8 weeks, including a 5-class image classification CNN with >90% validation accuracy and a YOLOv5 object detection model served via Flask API.

Member | MLRIT CIE

July 2022 - October 2023

Web Technologies, Figma, Photoshop

- Shipped 3+ frontend features for the MLRIT CIE website using Wordpress, HTML, CSS, and JavaScript, improving navigation and event visibility for 500+ student users.
- Mentored 100+ students in project development and technical fundamentals.
- Coordinated technical events including Inventron and Metaloop.

CERTIFICATIONS

Introduction to Machine Learning | Issued by NPTEL

Introduction to Artificial Intelligence | Issued by LinkedIn Learning

Python | Issued by HackerRank

Java | Issued by HackerRank

LANGUAGES

English - Fluent | **Hindi** - Fluent | **Telugu** - Native